

Science

Science | Class 10 - A | Due 2026-05-29 | 28 marks

Student Information

Name: _____

Roll Number: _____

Section: _____

Section A

Attempt all questions.

1. Which of the following is the correct formula for calculating the kinetic energy of an object?

- A. $KE = m \times v$
- B. $KE = \frac{1}{2} \times m \times v^2$
- C. $KE = m \times g \times h$
- D. $KE = F \times d$

Easy | 1 mark

2. What happens to the pH of a solution when an acid is added to it?

- A. pH increases
- B. pH decreases
- C. pH remains the same
- D. pH becomes neutral

Moderate | 1 mark

3. Which of the following elements has the highest electronegativity?

- A. Sodium (Na)
- B. Chlorine (Cl)
- C. Oxygen (O)
- D. Carbon (C)

Moderate | 1 mark

4. In the periodic table, elements are arranged in order of increasing:

- A. Atomic mass
- B. Atomic number
- C. Atomic radius
- D. Valence electrons

Hard | 1 mark

5. Explain the difference between renewable and non-renewable energy sources with examples.

Easy | 2 marks

6. Describe the process of photosynthesis and write the chemical equation involved.

Moderate | 2 marks

7. State Newton's three laws of motion and provide an example for each.

Hard | 2 marks

8. Discuss the structure of an atom, including the location and charge of protons, neutrons, and electrons. Explain how the arrangement of electrons in different energy levels determines the chemical properties of elements.

Moderate | 5 marks

9. Compare and contrast mitosis and meiosis. Include in your answer the purpose, number of divisions, and the genetic outcome of each process.

Hard | 5 marks

10. A local factory is discharging untreated waste into a nearby river, affecting the aquatic ecosystem. As an environmental scientist, analyze the potential impacts of this pollution on the river ecosystem and propose a comprehensive plan to mitigate these effects. Your answer should include both immediate and long-term solutions.

Moderate | 8 marks